SPETC v10 — User Guide

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Abstract

This guide explains how to install, configure and operate SPETC, the spectro-photometric exposure time calculator, from the first launch to photometric and spectroscopic planning, results interpretation and CSV export. No knowledge of the internal physics is required; the companion document *Physics of the ETC* covers the models and formulae.

Contents

1 Getting started **2**
- 1.1 Requirements and installation . . . 2
- 1.2 First launch . . . 2
- 1.3 A one-minute first calculation . . . 2

2 The main window **2**
- 2.1 Observation, site and target . . . 2
- 2.2 Telescope, detector and conditions . . . 3
- 2.3 Calculation settings . . . 3
- 2.4 Filter and template selection . . . 3
- 2.5 Run actions . . . 3

3 Photometry step by step **3**
- 3.1 Point sources . . . 3
- 3.2 Extended sources (galaxies, nebulae, planets) . . . 4
- 3.3 Target S/N mode and the stack planner . . . 4
- 3.4 Differential photometry (variable stars, exoplanets) . . . 4

4 Spectroscopy step by step **4**
- 4.1 Slit spectroscopy . . . 4
- 4.2 Slitless spectroscopy and the Star Analyser . . . 5
- 4.3 Getting R from your hardware . . . 5
- 4.4 Reading the spectroscopic output . . . 5

5 Results, plots and export **5**
- 5.1 The Results window . . . 5
- 5.2 Scientific plots . . . 6
- 5.3 Practical notes . . . 6

6 Troubleshooting **6**

7 Command-line tools 6
7.1 Calibrating your own filters: make_filter_profile.py 6
7.2 Importing template spectra: add_template.py 7
7.3 Headless runs: spetc_batch.py 8

1 Getting started

1.1 Requirements and installation

SPETC needs Python 3.10 or later with Tk support. Install the dependencies and start the program from the distribution directory:

```
python3 -m pip install -r requirements.txt
python3 etc_gui.py
```

The `data/` directory shipped with the program must stay beside `etc_gui.py`; it contains the stellar template catalogue (`interpola.db.csv` and the CALSPEC/BPGS spectra), the default detector QE curve (`qe.dat`), the SVO filter profiles with their list (`filters.list`), the optional `BLANK.xml` unfiltered passband and the optional Earth-atmosphere FITS table. The data directory can also be pointed elsewhere from the Stellar template selector panel.

1.2 First launch

On start-up SPETC restores the last saved configuration from `etc_user_config.json` (created automatically; the default preset is Asiago/Cima Ekar) and the most recently used instrument profile, if any. The configuration is re-saved after every successful calculation and on normal exit, so the program always reopens as you left it.

1.3 A one-minute first calculation

1. Pick a site from the Observatory selector (or type latitude, longitude, elevation).
2. Set the date and UT time, or leave the current time.
3. Type target coordinates (decimal or sexagesimal), or type a name and press Resolve name (needs internet; nothing else ever does).
4. Select an observing filter and a stellar template; enter the target magnitude.
5. Press Run ETC. The Results window opens with the numbers; the plot window shows S/N versus time, altitude, and the sky path.

2 The main window

The window is organised in five columns, left to right.

2.1 Observation, site and target

Site selector (built-in observatories merged with your persistent `etc_sites.json`; Save current site and Import site file manage it), geographic coordinates and elevation, date and UT time of the planned observation, the observing-interval selector, the time zone (IANA name with DST, e.g. `Europe/Rome`, or a fixed UTC offset), and the target: RA/Dec fields (decimal degrees or `hh:mm:ss` / `dd:mm:ss`) with the optional SIMBAD Resolve name button. Running the ETC itself never uses the network: only the coordinates in the fields count.

2.2 Telescope, detector and conditions

Telescope diameter, central obstruction, focal length and scalar optics throughput; optional calibrated instrument-response and slit resolving-power curves (two-column files with an explicit wavelength unit); detector pixel size, gain, full well, ADC bit depth, read noise and dark current; the **sensor type** (`mono`, or `osc` with the R/G/B channel: aperture rates and pixel counts scale by the Bayer fill fraction, saturation stays per physical pixel; use the channel's effective QE curve); a **CMOS gain table** (a text file with one row per gain setting: setting, e^{-}/ADU, read noise, full well — published by camera makers and measured on PhotonsToPhotos): pick the setting and the three detector fields fill consistently; seeing FWHM; the sky-background mode (`ing` = physical night/twilight/day model, `fixed_ab` = your own observed sky surface brightness, `sqm` = your **zenith SQM reading** in V mag/arcsec^2, with a **Bortle class** preset that fills a typical value; band colours and the Moon model are applied on top); and the **PSF model** (`gaussian` or `moffat` with its β; Moffat reproduces the wings of real star images and is recommended). Save/Load instrument profile stores the whole column as a portable JSON with its QE and response curves.

2.3 Calculation settings

Photometry or spectroscopy; exposure time; optional **target S/N** (when set, the ETC returns the required exposure instead, flagging saturation — the solver includes scintillation, so bright-star requests are honest); the **stack sub-exposure** preference (0 = automatic) feeding the stack planner, whose result appears in the label below after each run. The photometry panel adds the aperture radius, the **source geometry**: `point`, or `extended` with the source area in arcsec^2 for galaxies, nebulae and planets (the magnitude you enter stays the *total* integrated magnitude), and the optional **comparison-star magnitude** for differential photometry — the mmag-per-frame precision appears in the label below. The spectroscopy panel holds the resolving power, slit width, extraction height, sampling, the slitless parameters (Sect. 4), the wavelength range and the S/N reference wavelength, the **slit orientation** (`parallactic` or `fixed`), and the **spectrograph geometry helper**: enter grating lines/mm and collimator/camera focal lengths and press Compute R from geometry — the Littrow grating equation fills the R field (which stays editable) and reports whether you are slit- or seeing-limited.

2.4 Filter and template selection

Two independent filters: the **reference** filter, in which your magnitude is expressed (Vega or AB), and the **observing** filter, physically in the beam for both photometry and spectroscopy (spectroscopy defaults to `BLANK`, the unfiltered response). The panel previews the observing transmission curve with its pivot wavelength, width and zero point; the template selector searches the catalogue by name and previews the selected flux distribution; Validate V and B−V checks the template's synthetic colours against the catalogue.

2.5 Run actions

Run ETC, the live observatory status (Sun/Moon altitude, current airmass of the target), and Exit (which saves the configuration).

3 Photometry step by step

3.1 Point sources

1. Set Source geometry to `point`.

2. Choose the aperture radius. As a rule of thumb $1.5\times$ the seeing FWHM captures $\sim$95% of a Moffat profile while limiting sky noise; the ETC applies the exact encircled-energy correction, so any radius gives a correct answer.

3. Enter the target magnitude and its reference filter and system (Vega default; AB available).
4. Run. The S/N-versus-time curve shows the whole night; the red marker is your selected time.

For bright stars do not be surprised that the S/N stops growing with brightness at a few $\times 10^3$: atmospheric scintillation sets a ceiling per exposure that no photon count can beat. Stack short frames to go beyond it.

3.2 Extended sources (galaxies, nebulae, planets)

Set Source geometry to `extended` and give the source area in arcsec^2 (e.g. a $2' \times 1'$ galaxy ≈ 7200; Jupiter ≈ 1500 at opposition). Keep entering the *integrated* magnitude: the ETC spreads it uniformly over the stated area, so the aperture collects the corresponding area fraction and pixel saturation is judged on the surface brightness, which is what actually saturates on a resolved object. The approximation is intended for sources clearly larger than the seeing disc.

3.3 Target S/N mode and the stack planner

Enter a value in Target S/N to invert the calculation: the ETC solves for the exposure in closed form, including read noise *and scintillation* (bright-star requests would otherwise undershoot badly), and the time-series plot switches to required exposure versus time. The **stack planner** then answers "how many subs?" in the label under CALCULATION: $N \times t_{\text{sub}}$ (the sub capped by pixel saturation or your entered preference), the total time, the read-noise penalty relative to one ideal long exposure, and the frame length above which the sky background dominates read noise. Without a target S/N the label still shows the saturation cap and the sky-limited frame length as guidance.

3.4 Differential photometry (variable stars, exoplanets)

Enter the magnitude of your comparison star in Comparison star mag: after the run the PHOTOMETRY label (and the Assumptions/outputs tab) reports the differential precision in **millimagnitudes per frame**, combining both stars' full noise budgets (each including scintillation, treated as uncorrelated — conservative). "3.2 mmag per 60 s frame" is directly the number AAVSO and transit observers plan with: a 12 mmag transit needs several frames per point at that precision.

4 Spectroscopy step by step

4.1 Slit spectroscopy

1. Mode `slit`; set the resolving power R (or load a measured slit-width/R curve, which then overrides it), the slit width and the extraction height in arcsec, and pixels per resolution element.
2. Set the wavelength range and the S/N reference wavelength: the time-series evaluates that wavelength bin at every time sample.
3. Choose the **slit orientation**. `parallactic` assumes you rotate the slit to the parallactic angle (reported in the Results window for every time), avoiding atmospheric-dispersion losses. `fixed` applies the worst-case per-wavelength loss — expect tens of per cent missing at the blue end at airmass 1.5–2.
4. Run. The plot window shows the detected source rate and S/N per resolution element, with a manual time slider through the night.

4.2 Slitless spectroscopy and the Star Analyser

Mode `slitless`. Two ways to describe the disperser:

- **Star Analyser 100/200 (recommended)**: enter the groove density (100 or 200 lines/mm) and the grating-to-sensor distance in mm; the dispersion in Å/pixel follows from the physics (an SA100 at 42 mm on 4.63 µm pixels gives $\simeq 11$ Å/pixel). Optionally set the first-order grating efficiency (~ 0.5 is realistic).
- **Manual**: leave the groove density at 0 and type the dispersion in Å/pixel directly.

The resolution element is set by seeing, the intrinsic grating LSF and atmospheric dispersion; with 2–3$''$ seeing expect $R \sim$ 100–200 regardless of exposure. The cross-dispersion extraction height is the only aperture; there are no slit losses. Slitless results are flagged *experimental* until validated on a calibrated instrument.

4.3 Getting R from your hardware

If you know your grating (lines/mm) and the collimator and camera focal lengths but not R: fill the three geometry fields and press Compute R from geometry. The Littrow grating equation, your slit width, the telescope focal length and the seeing give R, the resolution element in Å and the dispersion in Å/mm, and state whether the slit or the seeing limits you. The value lands in the R field and can be edited freely afterwards.

4.4 Reading the spectroscopic output

All spectroscopic counts and S/N are **per resolution element**, not per pixel. The Results table lists, per wavelength: the element width, the effective R, source and sky rates, S/N, σ**(EW)** in mÅ, peak electrons and ADU of the brightest pixel, the saturation flag and the maximum unsaturated exposure.

The line criterion. σ(EW) is the Cayrel (1988) equivalent-width uncertainty: a spectral line is securely measurable when its expected equivalent width exceeds about $3\,\sigma$(EW). Example: σ(EW) $= 15$ mÅ at Hα means the 50–100 mÅ variations of a Be star are detectable, while 20 mÅ photospheric lines are marginal. The value at the S/N reference wavelength is shown in the PHOTOMETRY/summary label and in the Assumptions/outputs tab.

5 Results, plots and export

5.1 The Results window

Three tabs:

- **Selected-time result**: the numerical table for the chosen time (magnitude row in photometry; per-wavelength rows in spectroscopy), showing *every* quantity the engine produced: rates, S/N, σ(EW) [mÅ] for spectroscopy, the scintillation and ADC-quantization noise in electrons, pixels in the aperture, peak electrons/ADU, saturation cause, maximum unsaturated exposure, the standard and instrumental magnitudes and the sky brightness used. Save result CSV exports the full table.
- **Time series**: one row per time sample with UTC and local time, MJD, elevation, azimuth, **parallactic angle**, airmass (Pickering, refraction-corrected), S/N or required exposure, **sky surface brightness** actually used, band-effective extinction and its per-airmass coefficient, the absorption-adjusted apparent magnitude, saturation flag, peak electrons and maximum unsaturated exposure. This table is the CSV export.

- **Assumptions / outputs**: a plain-language statement of every physical assumption of the current run — sky model and its components (including the SQM value in `sqm` mode), PSF model, the CMOS gain-table setting in use, the spectrograph geometry, noise terms, saturation policy — plus the selected-time altitude, airmass, parallactic angle and sky brightness, followed by a **SELECTED-TIME OUTPUTS** block: S/N, scintillation noise in electrons and as a percentage of the source, ADC noise, peak-pixel and saturation numbers, predicted magnitudes, the σ(EW) line criterion, the stack plan and the differential-photometry precision.

5.2 Scientific plots

The native Matplotlib window provides pan/zoom/save, manual axis limits (time axes accept ISO timestamps), autoscale per panel, and in spectroscopy a manual slider that walks the sampled times of the observing interval, updating the spectrum panels and the markers. The sky-path panel shows target, Sun (daytime only) and Moon (only above the horizon) in altitude–azimuth (N=0°, E=90°).

5.3 Practical notes

- The Moon's effect is included automatically in `ing` sky mode whenever it is above the horizon; within $\sim 30°$ of a bright Moon expect the sky several magnitudes brighter.
- In `fixed_ab` mode the entered sky brightness is used as-is at every time — use it when you have a measured value.
- Airmass and all results are deliberately unavailable below $5°$ altitude.
- Save an **instrument profile** once your telescope/detector column is right; it restores automatically at start-up and travels with its QE/response files.

6 Troubleshooting

If a run aborts with a Python traceback, the full text is shown in the error dialog: it names the offending input explicitly and is the first thing to attach to a bug report, together with `etc_user_config.json`.

7 Command-line tools

7.1 Calibrating your own filters: `make_filter_profile.py`

Amateur filters (Astrodon, Astronomik, Baader, Optolong RGB and narrowband) are not in the SVO database, but a transmission curve is all that is needed — the zero points are computed, not measured:

```
python3 make_filter_profile.py Astrodon.G astrodon_g.dat nm
```

reads a two-column wavelength/transmission file (0–1 or percent, auto-detected; unit Angstrom, nm or um), and writes `Astrodon.G_Vega.xml` and `Astrodon.G_AB.xml` into `data/filters/` in full SVO format: complete VO annotation, all characterising wavelengths, FWHM, effective width, the mean solar flux, and a Vega zero point computed synthetically from the bundled CALSPEC Vega spectrum — the same convention SVO uses (validated to $\lesssim$1% against the shipped SVO profiles). Add the two printed lines to `data/filters.list` and the filter appears in the selector on the same photometric scale as the professional passbands.

Table 1: Common messages and their resolution.

Message / symptom	Cause and fix
"…covers X–Y Å, but the requested calculation needs…"	A curve (template, QE, filter, atmosphere) does not cover the requested wavelengths. Narrow the range or load data with wider coverage; the ETC never extrapolates silently.
"Target altitude is below 5 deg"	The selected time puts the target too low. Move the time or check the coordinates; the visibility plot shows when the target rises.
"The template has no positive flux in the selected magnitude bandpass"	Reference filter outside the template's spectral range (e.g. K band with an optical-only spectrum). Choose a filter inside the template coverage.
Required exposure flagged as saturating	The target S/N needs a longer exposure than the brightest pixel allows. Use the reported maximum unsaturated exposure and stack frames.
S/N of a bright star lower than expected	Scintillation ceiling (physical, not an error). Stack short exposures; larger aperture and lower airmass raise the ceiling.
"Resolve name" fails	SIMBAD needs internet; type coordinates manually — nothing else requires the network.
Filters missing at start-up	`data/filters.list` not found or empty. Each logical filter needs its `_Vega` and `_AB` XML profiles; `BLANK.xml` in `data/filters/` enables the unfiltered response.
Slit width outside the calibrated range	A measured slit-R curve is loaded and the requested width lies outside it. Change the width or remove the curve.

7.2 Importing template spectra: `add_template.py`

Template spectra can be two-column ASCII *or* CALSPEC/STScI-style FITS tables (`WAVELENGTH`/`FLUX`; Angstrom, nm or micron units read from `TUNIT`). To add a single file or an entire directory — e.g. the whole CALSPEC archive folder:

```
python3 add_template.py current_calspec/
python3 add_template.py my_spectrum.fits "HD 12345" G2V
```

For each spectrum the tool copies the file under `data/imported/`, computes the synthetic Vega V magnitude (the catalogue `mv0`) and $B-V$ against the shipped Bessell profiles, cleans CALSPEC filename suffixes for the display name (`alpha_lyr_stis_011` $\rightarrow$ `alpha_lyr`), reports the V-band coverage, and appends the row to `data/interpola.db.csv` — the file the GUI template list is built from. Press Reload catalog (or restart) and the new templates appear in the selector. Surface-brightness and arbitrary-flux files (planet atlas, supernova templates) are fully usable: the ETC rescales every template to your entered magnitude, so only the spectral shape and the computed `mv0` matter. Files that do not fully cover the V band are flagged; use V as the reference filter for those.

7.3 Headless runs: `spetc_batch.py`

```
python3 spetc_batch.py examples/batch_photometry.json out/
```

runs the same engines without the GUI from a JSON run description (two commented examples ship in `examples/`) and writes the full result CSV plus a self-contained **one-page HTML summary**: configuration, sky, key numbers (σ(EW) for spectroscopy, saturation, maximum unsaturated exposure), the stack plan, and an embedded S/N figure (S/N versus exposure for photometry; S/N versus wavelength for spectroscopy). Batch sky modes are `fixed_ab` and `sqm`; the Moon and twilight terms need the GUI's time machinery and are not applied, as the summary footer states. Useful for scripting parameter studies and for archiving a session plan.